ADVANCED TASK MANAGER WEB APP TESTING

Software Testing CCSW 323

PREPARED BY :

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introduction

- *Project: Software Testing of a Web Application*
- *Apply testing models and coverage criteria*
- *Design and execute test cases*
- *Use automation tools (Selenium & JUnit)*
- *Evaluate system functionality and reliability*

SOFTWARE UNDER TEST

The To-Do List Web Application is a simple web-based system that helps users manage and organize their daily tasks efficiently.

- **Built using:** HTML, CSS, JavaScript
- **System Size:** Approximately 320 Lines of Code (LOC)
- **Goal:** To ensure that the system meets the required specifications, supports user needs, and prevents invalid or incorrect user inputs to maintain accurate operation.

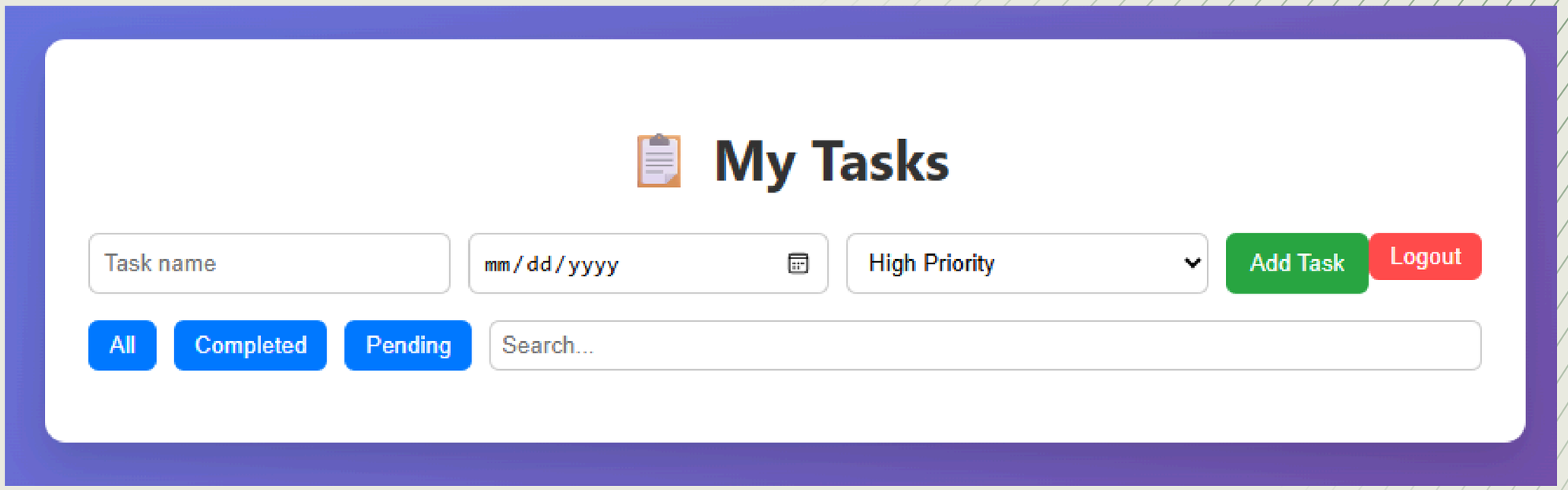
SOFTWARE UNDER TEST

Application Features

- User login system
- Add new tasks
- Edit existing tasks
- Mark tasks as completed
- Delete tasks
- Search and filter tasks

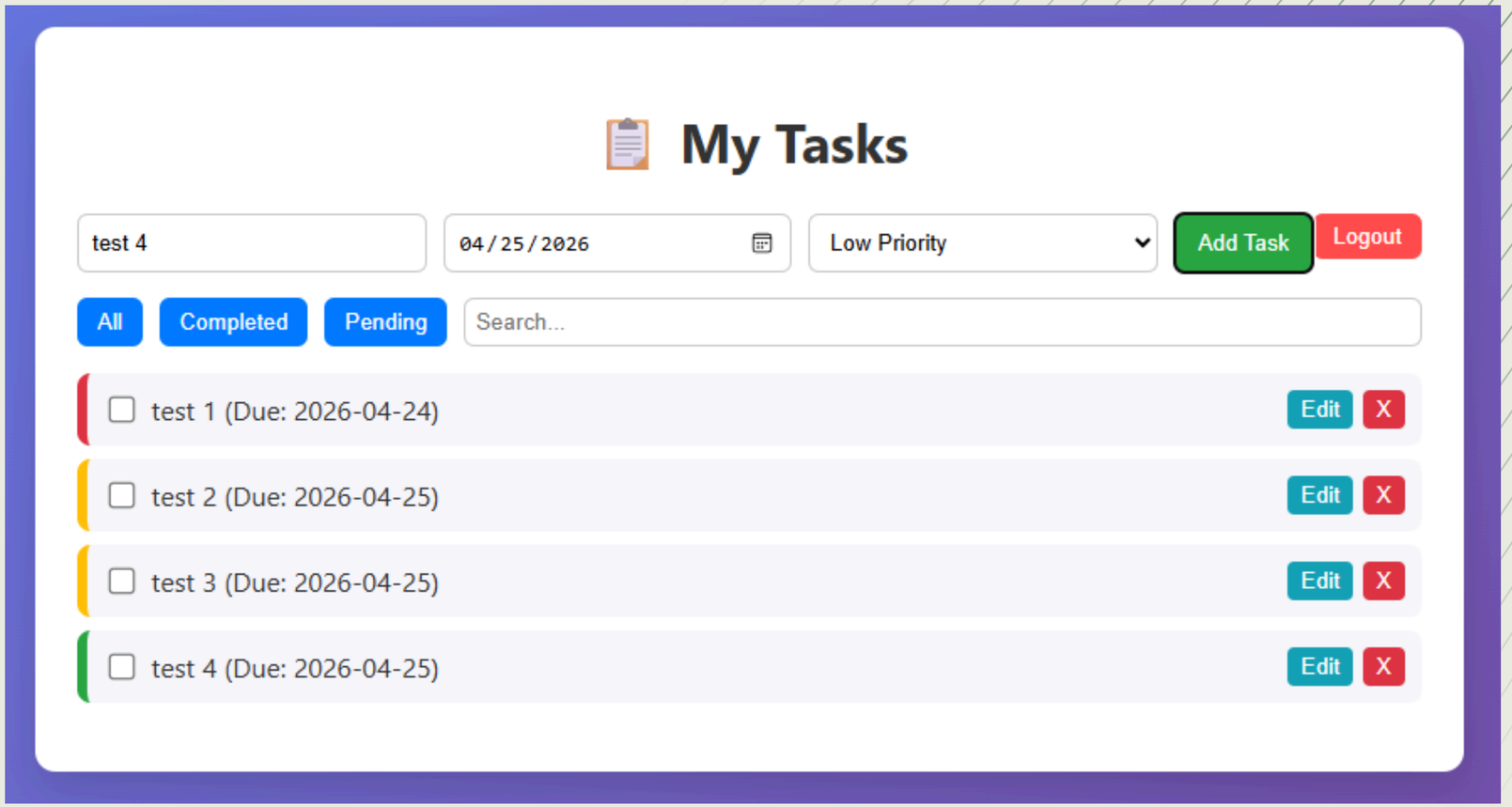
SOFTWARE UNDER TEST

Main page after successful login.



SOFTWARE UNDER TEST

After adding three different priority tasks.



TESTING MODELS

1. Graph-Based Testing

- Test user actions (flow)
- Example: add → complete → delete

2. Input Space Testing

- Test different inputs
- Example: empty, wrong, valid

COVERAGE CRITERIA

GRAPH-BASED TESTING

NC (Node Coverage)

- Test all system states
- Add / Edit / Delete / Complete

EC (Edge Coverage)

- Cover each transition
- Test all system flows

COVERAGE CRITERIA

INPUT SPACE

BCC (Base Choice Coverage)

- Start with normal case
- Change one input each time

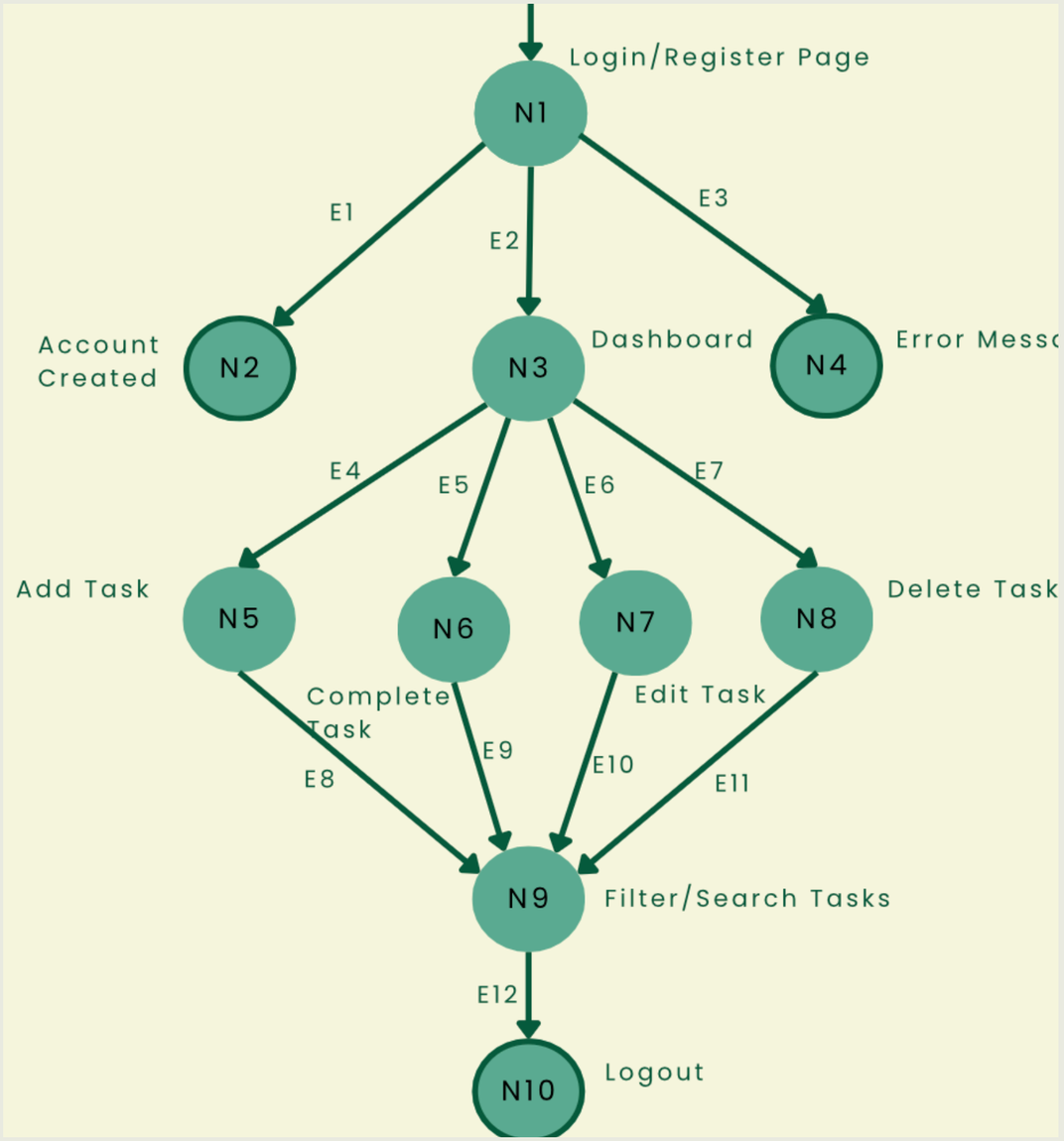
ECC (Each Choice Coverage)

- Test each input value once
- Cover all input types (valid, invalid, empty)

multiple base choice Coverage (MBCC)

- Use more than one normal case as a base
- Vary inputs around each base case to increase coverage

SYSTEM GRAPH MODEL



COVERAGE CRITERIA

Node Coverage (NC)

Test Case ID	Input Data	Node Covered	Expected Output	Actual Output
NC-01	Enter a new valid email and password, then click Create Account	N1, N2	Account is created successfully	Account created successfully
NC-02	Enter valid email and password, then click Login	N1, N3	User is redirected to dashboard	Login successful

COVERAGE CRITERIA

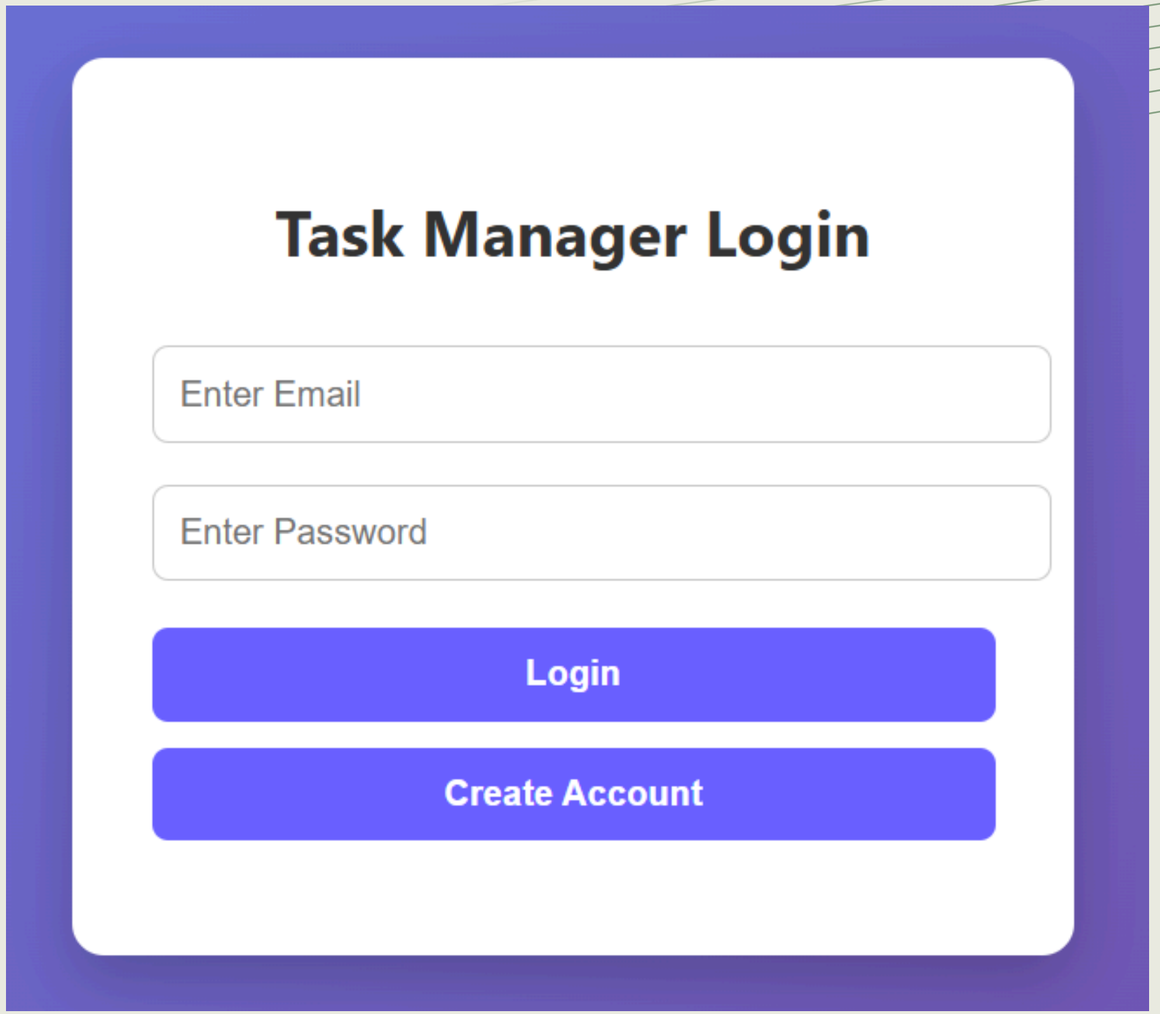
Edge Coverage (EC)

Test Case ID	Input Data	Edge Covered	Expected Output	Actual Output
EC-01	Enter a new valid email and password, then click Create Account	E1	Account created message shown	Account created successfully
EC-02	Enter valid email and password, then click Login	E2	User is redirected to dashboard	Login successful

COVERAGE CRITERIA

Base Choice Coverage (BCC)

Input: Task Manager Login



COVERAGE CRITERIA

Base Choice Coverage (BCC)

Characteristics: Email, Password

Blocks:
email: (valid , empty,wrong)
password: (valid, empty , wrong)

Abstract IDM:
A = [a1, a2, a3] , B = [b1, b2,b3]

Number of tests: 1(base)+2+2 = 5

BCC Test Cases:

base={a1 ,b1}
A {a2 ,b1}
A {a3 ,b1}
B {a1 ,b2}
B {a1 ,b3}

COVERAGE CRITERIA

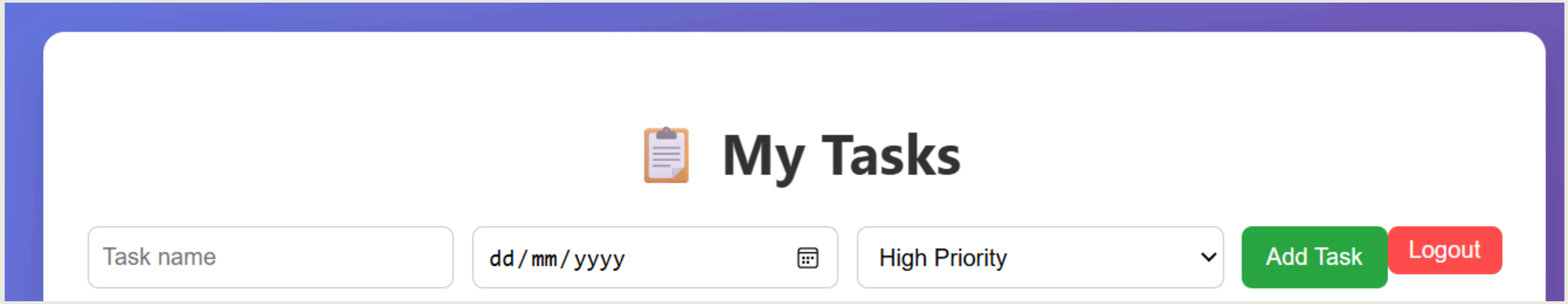
Base Choice Coverage (BCC)

Test Case ID	Input Data	Expected Output	Actual Output
BCC-01	Email: aa123@gmail.com, Password: 123456	User is redirected to the dashboard and login is successful	login is successful
BCC-02	Email: (empty), Password: 123456	System shows error message and login is not allowed	“Invalid credentials” message displayed

COVERAGE CRITERIA

Each Choice Coverage(ECC)

Input: add tasks



COVERAGE CRITERIA

Each Choice Coverage(ECC)

Characteristics: name task , due date ,priority

Blocks:
name taske : (valid , empty)
due date : (valid, empty)
priority:(high,medium ,low)

Abstract IDM:
A = [a1, a2] , B = [b1, b2] ,
C = [c1,c2,c3]

- **Number of tests: MAX(2,2,3) =3**

ECC Test Cases
{a1 , b1, c1}
{a2, b2, c2}
{a1, b1, c3}
NOTE :3 test cases were sufficient to cover all input characteristics and their corresponding blocks

COVERAGE CRITERIA

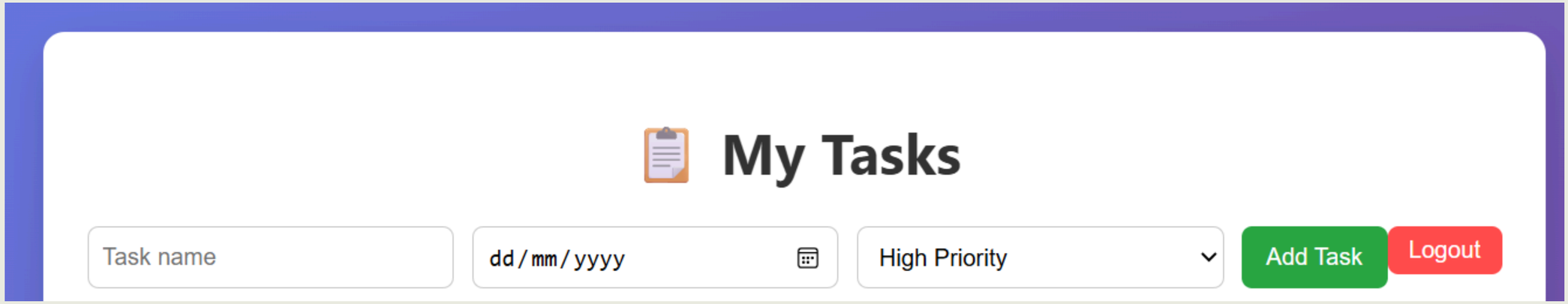
Each Choice Coverage(ECC)

Test Case ID	Input Data	Expected Output	Actual Output
ECC-01	Task: "study ch1", Date: 20/04/2026, Priority: High	Task is added successfully	Task added successfully
ECC-02	Task: "", Date: (empty)Priority: medium	Task NOT added, validation message shown	Validation message displayed

COVERAGE CRITERIA

multiple base choiceCoverage(MBCC)

Input: add tasks



COVERAGE CRITERIA

multiple base choiceCoverage(MBCC)

Characteristics: name task , due date ,priority

Blocks:
name taske : (valid , empty)
due date : (valid, empty)
priority:(high,medium ,low)

Abstract IDM:
A = [a1, a2] , B = [b1, b2] , C = [c1,c2,c3]

MBCC Test Cases
BASE1 {a1 , b1, c1}
BASE2 {a2, b2, c2}
Tests from a1, b1,c1:
a1, b1, c3,
Tests from a2,b2,c2:
a2, b2, c3,

COVERAGE CRITERIA

multiple base choice Coverage (MBCC)

base 1 & 2

Test Case ID	Input Data	Expected Output	Actual Output
MBCC-01	Task: "study ch1", Date: 20/04/2026, Priority: High	Task added successfully with high priority	Task added successfully
MBCC-02	Task: "", Date: (empty)Priority: medium	Task NOT added, validation message shown	Validation message displayed

COVERAGE CRITERIA

multiple base choice Coverage (MBCC)

test from base 1 &2

MBCC-03	Task: "Gym", Date: 22/04/2026, Priority: Low	Task added successfully with low priority	Task added successfully
MBCC-04	Task: "", Date: (empty)Priority: Low	Task NOT added, validation message shown	Validation message displayed

TESTING TOOL

They automate test execution, verify system behavior accurately, and improve testing efficiency and reliability.

TESTING TOOL

Selenium WebDriver
Used to automate browser actions and simulate user behavior.

JUnit
Used to execute test cases and verify expected results automatically.

TOOL USAGE

Selenium WebDriver Features

- Automates browser actions
- Supports multiple browsers
- Executes test cases automatically
- Simulates real user behavior

TOOL USAGE

Usage in Our Project

- Opened the web application
- Performed login operations
- Added tasks automatically
- Verified system responses

TOOL USAGE

JUnit Features

- Runs automated test cases
- Checks expected vs actual results
- Displays pass/fail status
- Generates test reports

TOOL USAGE

Usage in Our Project

- Executed automated test cases
- Verified system behavior
- Displayed test results
- Confirmed successful execution

JUnit

TOOL USAGE

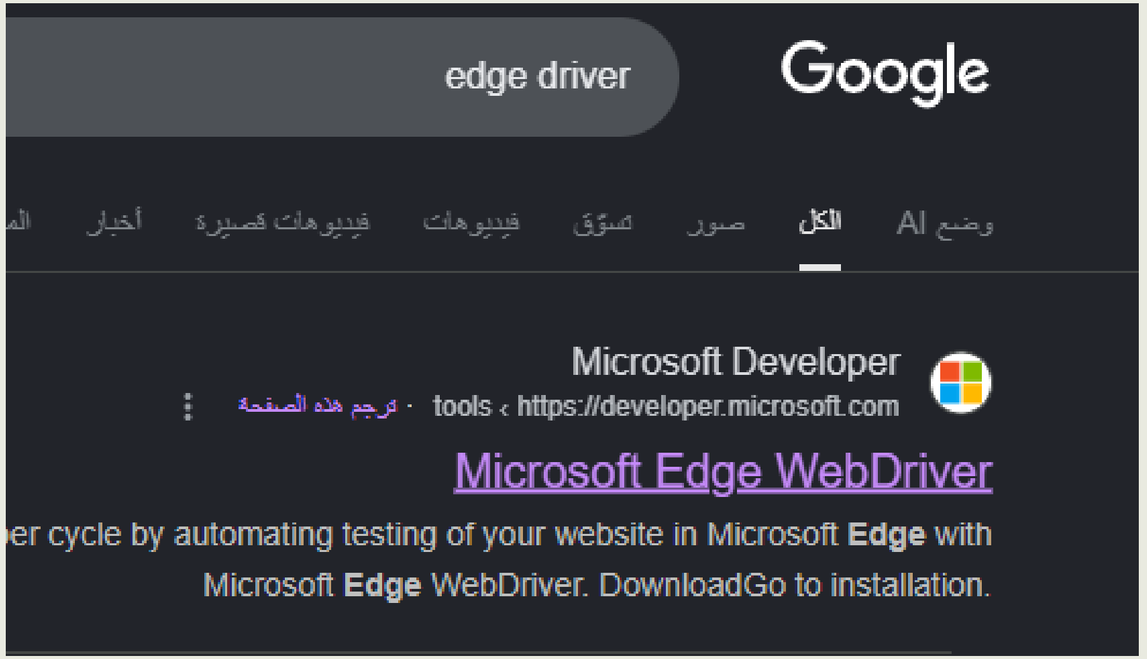
Installation

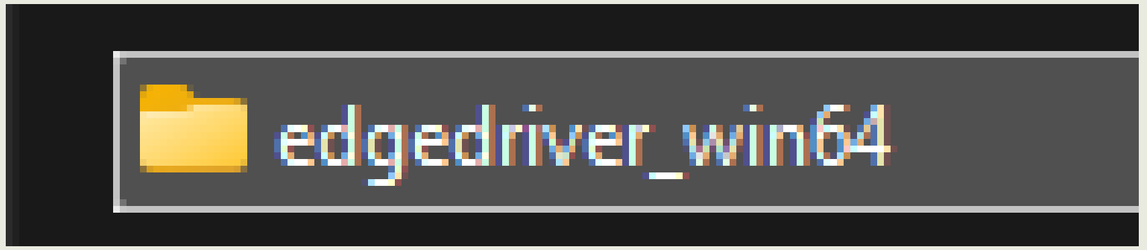
1. Download browser driver (e.g., EdgeDriver)
2. Add Selenium library to the project
3. Configure the driver path in the code
4. Add JUnit library to the project
5. Create a test class
6. Use @Test annotation to define test cases

TOOL USAGE

Installation

1. Download browser driver (e.g., EdgeDriver)

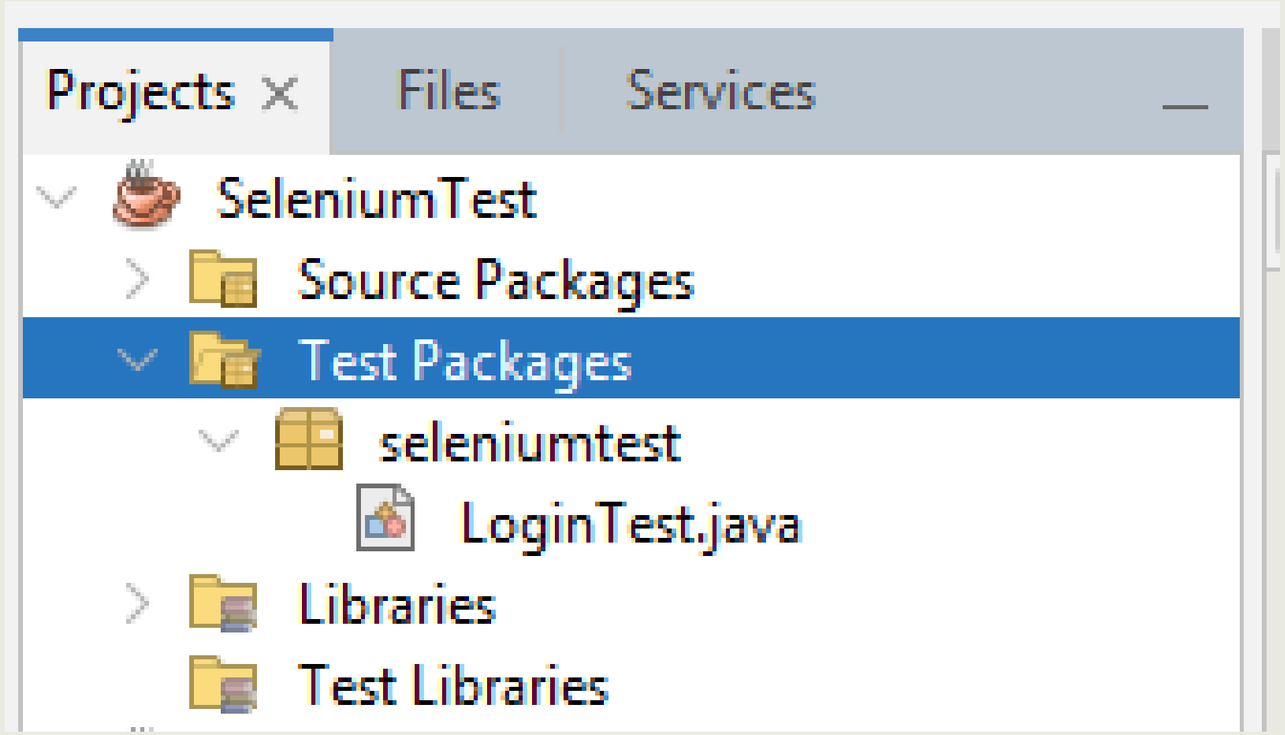




TOOL USAGE

Installation

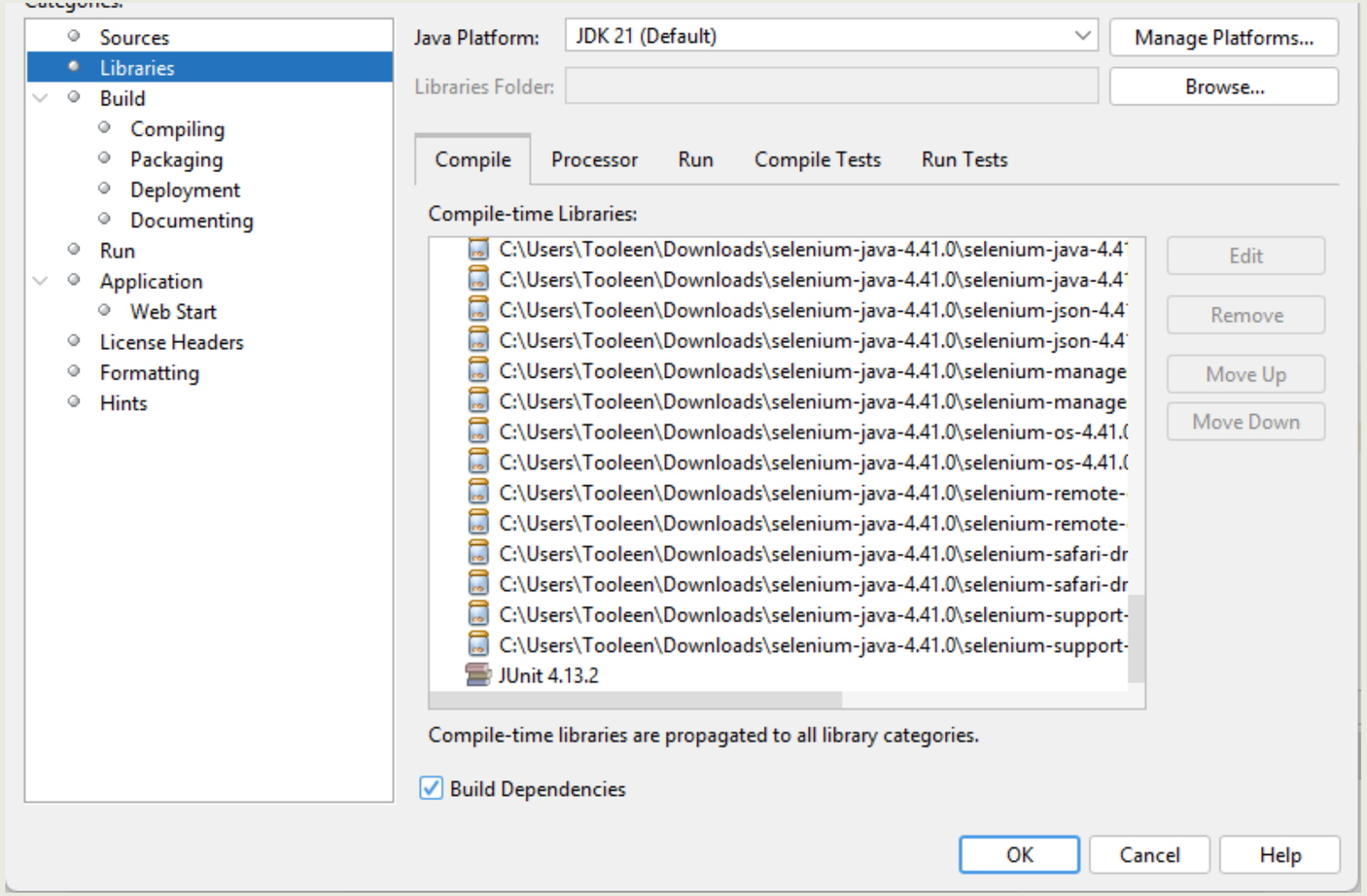
2. Create a test class



TOOL USAGE

Installation

3. Add JUnit and Selenium libraries to the project



TOOL USAGE

Installation

4. contains test methods using Selenium EdgeDriver and JUnit annotations to automate test execution.

```
5   public class LoginTest {
6
7       @Test
        public void testCreateAccount_LoginValidation_AddTask() throws InterruptedException {...98 lines }
6   }
```

TOOL USAGE

Example Test Case Execution Using Selenium and JUnit

TOOL USAGE

Test Case ID: BCC-02

Description:

This test case verifies the system behavior when the user attempts to login with an empty email field.

TOOL USAGE

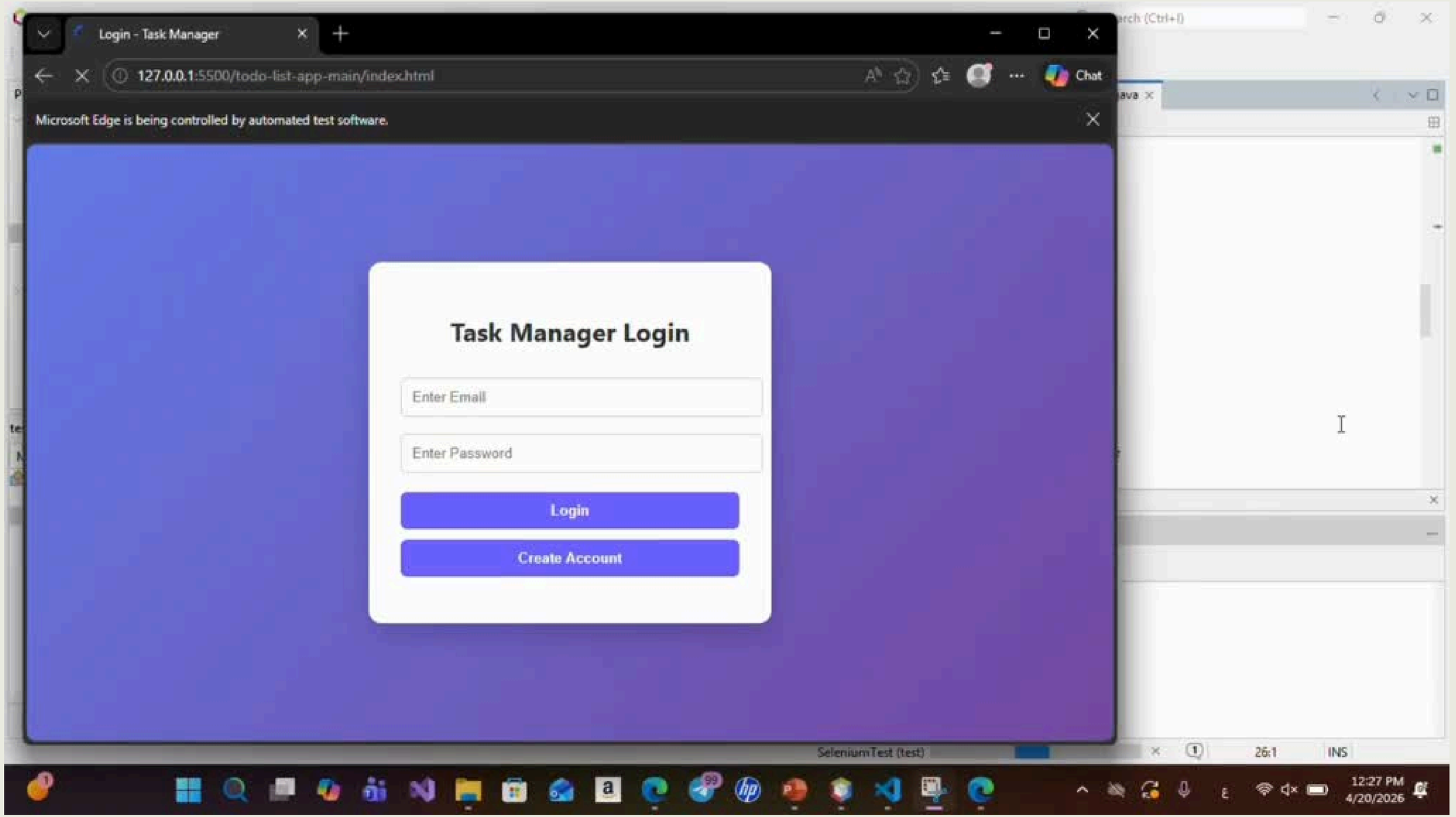
Test Case ID: BCC-02

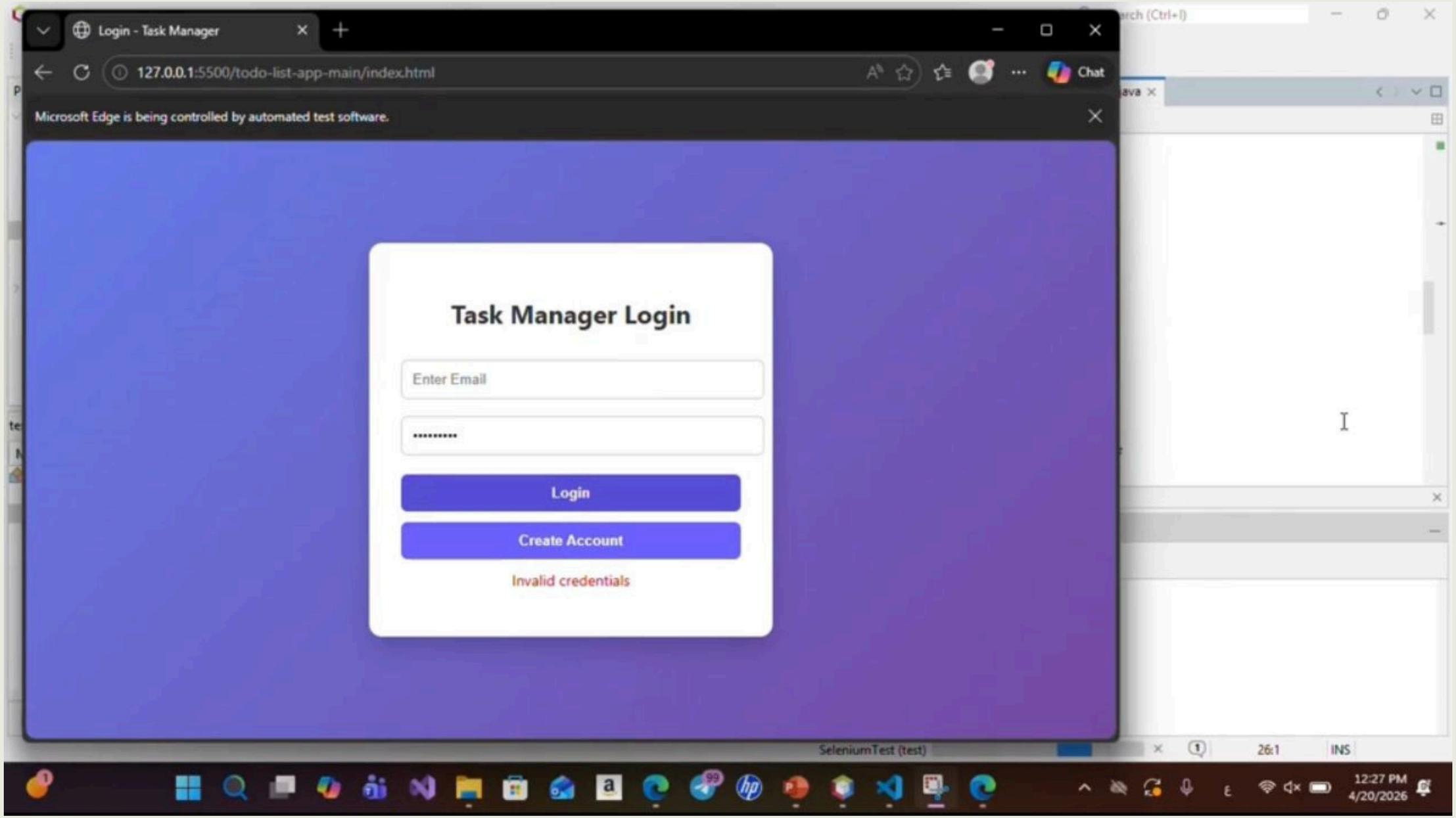
Description:
This test case verifies the system behavior when the user attempts to login with an **empty** email field.

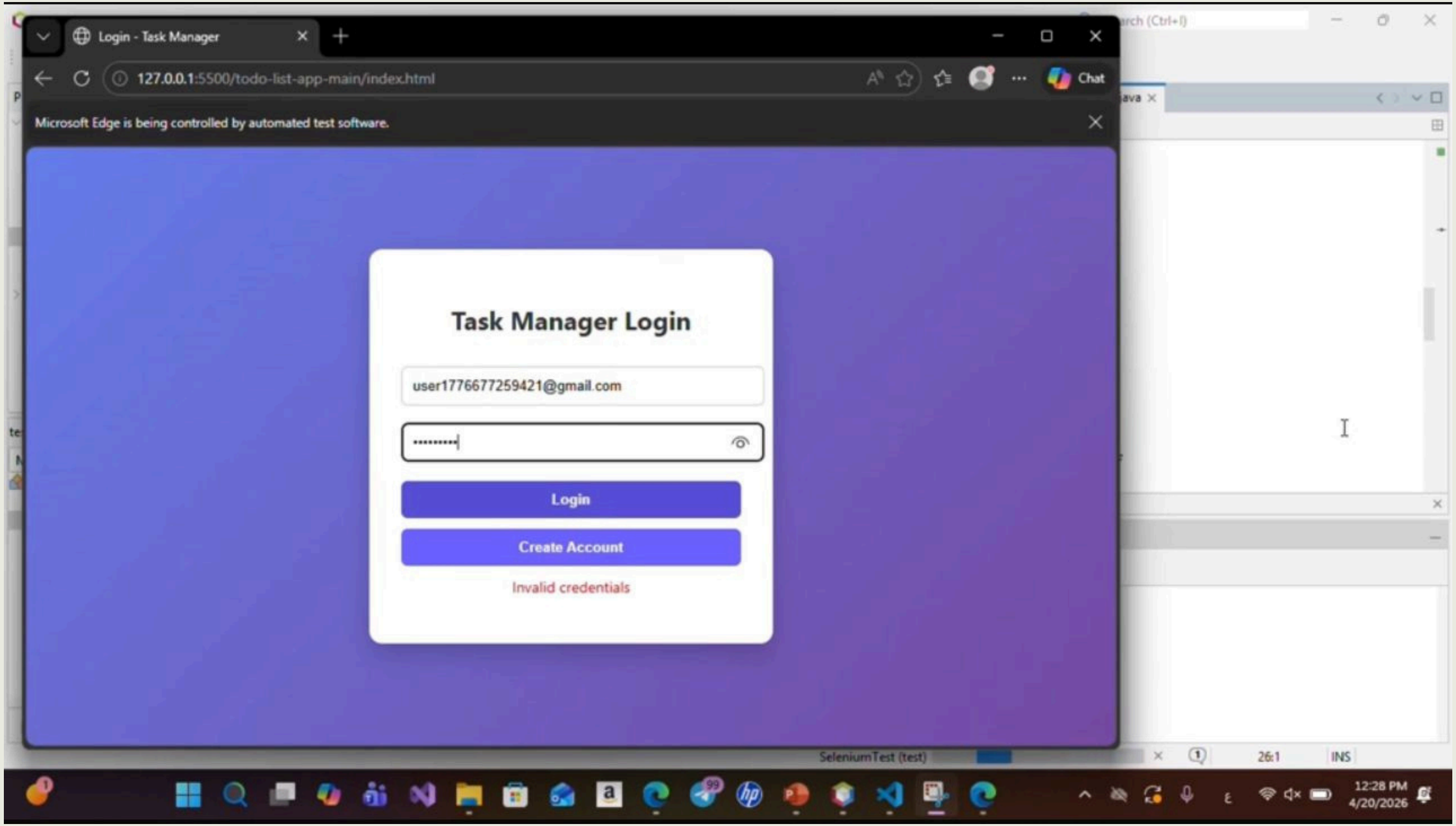
TOOL USAGE

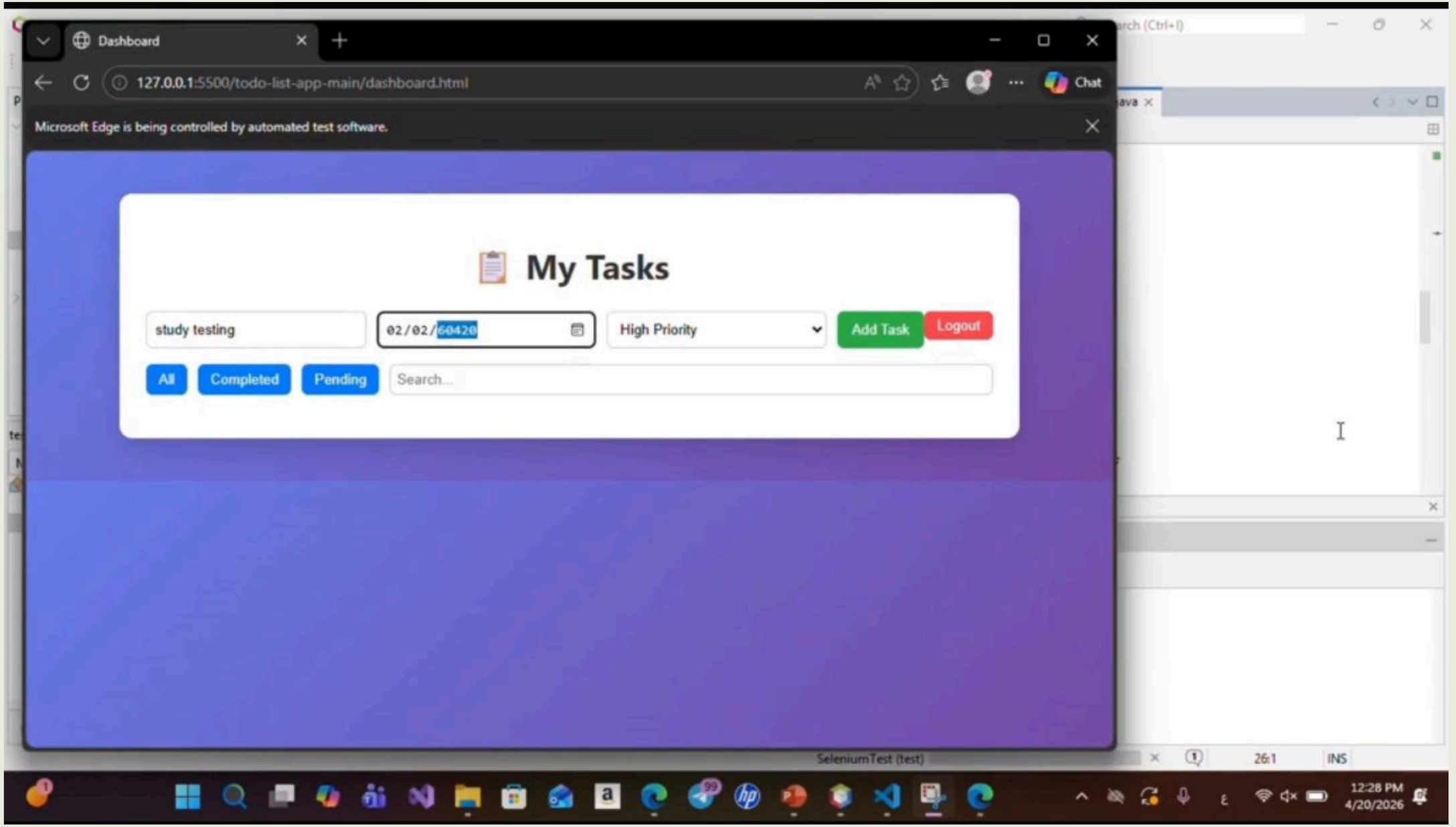
Test Case ID: ECC-01

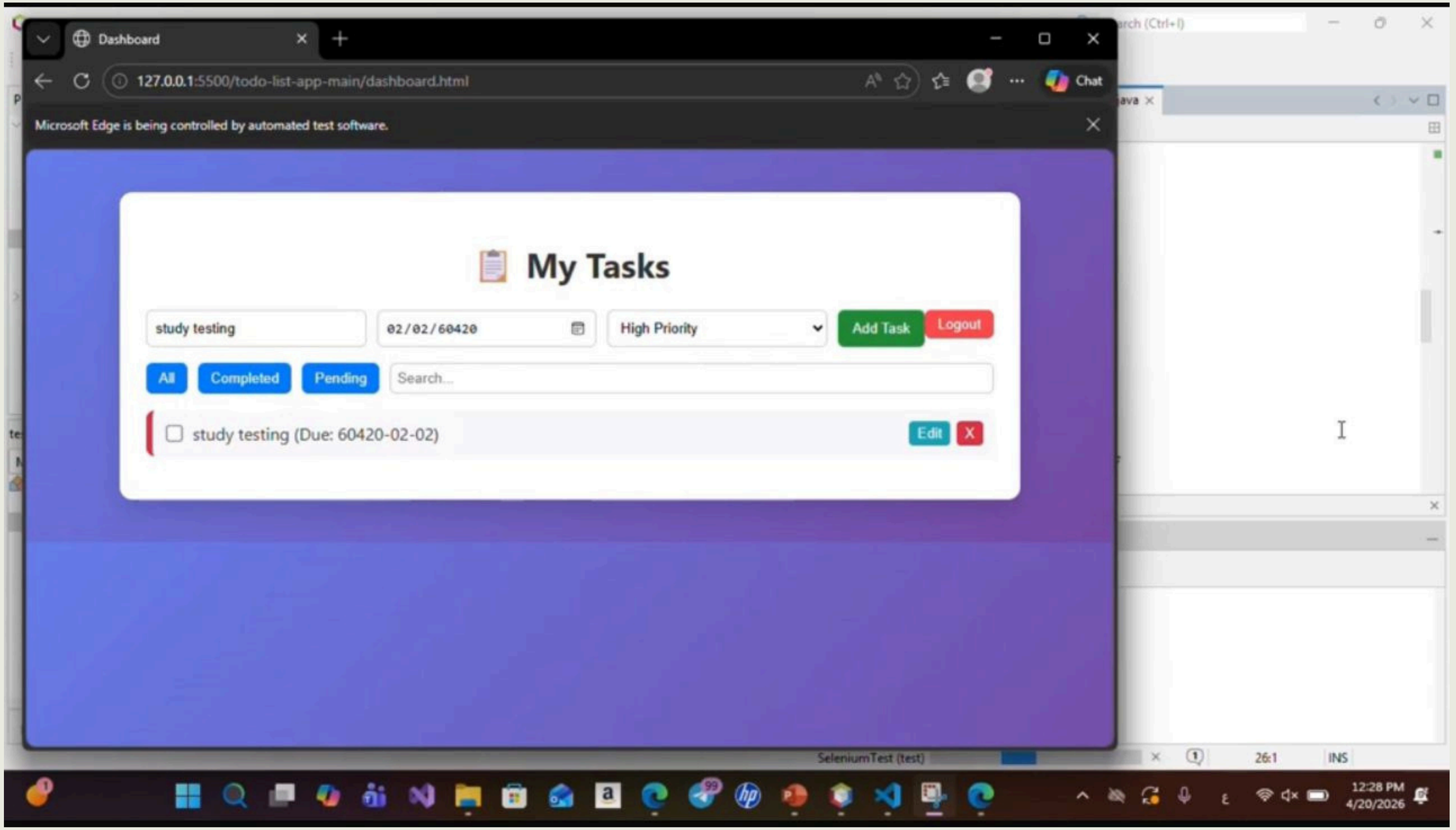
Description:
This test case verifies that the system allows the user to **add** a new task successfully using valid input data.

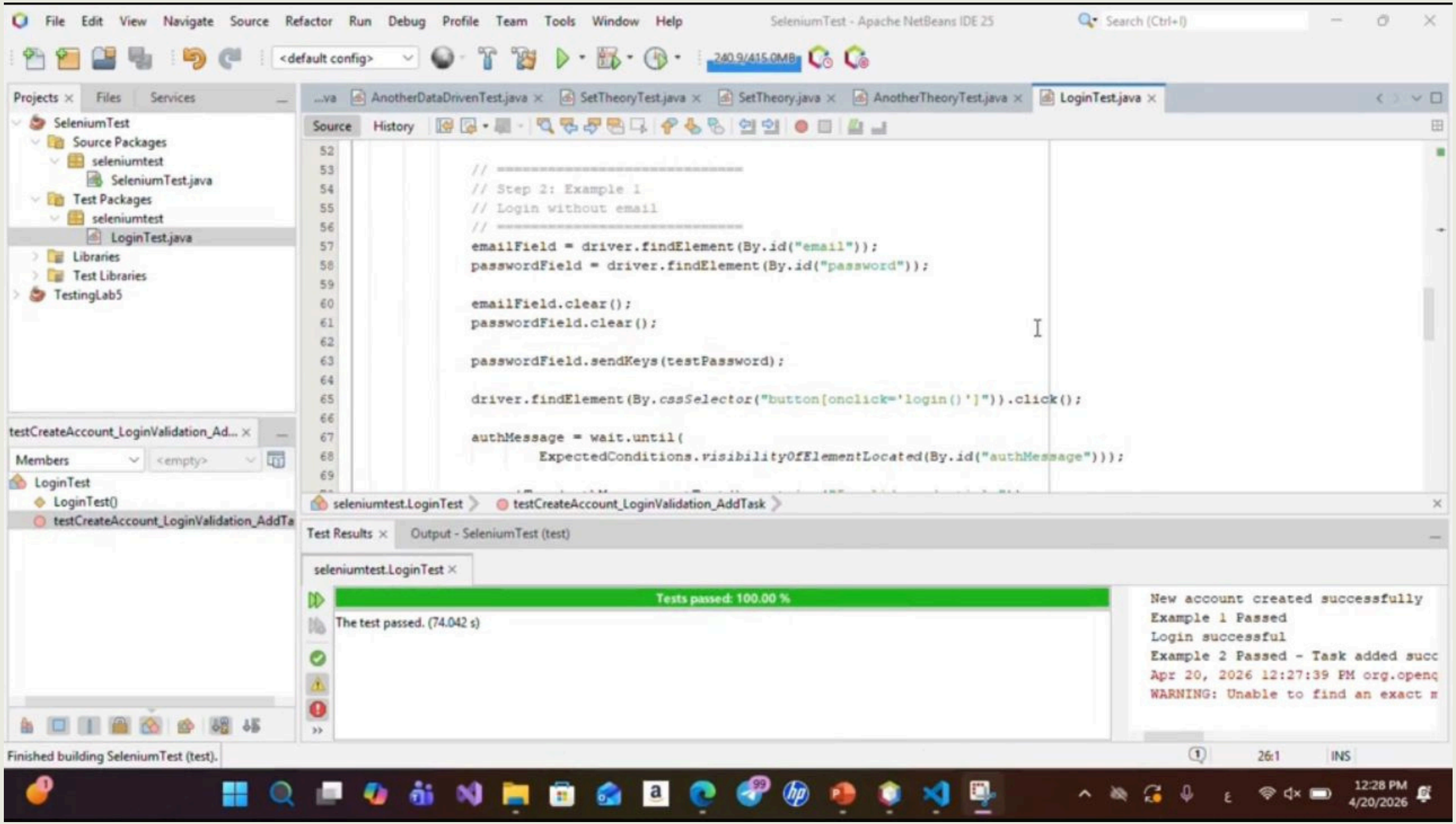

Login - Task Manager
127.0.0.1:5500/todo-list-app-main/index.html
Chat
Microsoft Edge is being controlled by automated test software.
Task Manager Login
Enter Email
Enter Password
Login
Create Account
SeleniumTest (test)
26:1
INS
12:27 PM
4/20/2026


Login - Task Manager
127.0.0.1:5500/todo-list-app-main/index.html
Chat
Microsoft Edge is being controlled by automated test software.
Task Manager Login
Enter Email
Login
Create Account
Invalid credentials
SeleniumTest (test)
26:1
INS
12:27 PM
4/20/2026


Login - Task Manager
127.0.0.1:5500/todo-list-app-main/index.html
Chat
Microsoft Edge is being controlled by automated test software.
Task Manager Login
user1776677259421@gmail.com
Login
Create Account
Invalid credentials
SeleniumTest (test)
26:1
INS
12:28 PM
4/20/2026


Dashboard
127.0.0.1:5500/todo-list-app-main/dashboard.html
Chat
Microsoft Edge is being controlled by automated test software.
My Tasks
study testing
02/02/60420
High Priority
Add Task
Logout
All
Completed
Pending
Search...
SeleniumTest (test)
26:1
INS
12:28 PM
4/20/2026


Dashboard
127.0.0.1:5500/todo-list-app-main/dashboard.html
Chat
Microsoft Edge is being controlled by automated test software.
My Tasks
study testing
02/02/60420
High Priority
Add Task
Logout
All
Completed
Pending
Search...
study testing (Due: 60420-02-02)
Edit
X
SeleniumTest (test)
26:1
INS
12:28 PM
4/20/2026


File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help
SeleniumTest - Apache NetBeans IDE 25
Search (Ctrl+I)
<default config>
240.9/415.0MB
Projects
Files
Services
SeleniumTest
Source Packages
seleniumtest
SeleniumTest.java
Test Packages
seleniumtest
LoginTest.java
Libraries
Test Libraries
TestingLab5
AnotherDataDrivenTest.java
SetTheoryTest.java
SetTheory.java
AnotherTheoryTest.java
LoginTest.java
Source
History
// ==========
// Step 2: Example 1
// Login without email
// ==========
emailField = driver.findElement(By.id("email"));
passwordField = driver.findElement(By.id("password"));
emailField.clear();
passwordField.clear();
passwordField.sendKeys(testPassword);
driver.findElement(By.cssSelector("button[onclick='login()']")).click();
authMessage = wait.until(
ExpectedConditions.visibilityOfElementLocated(By.id("authMessage")));
testCreateAccount_LoginValidation_Ad...
Members
<empty>
LoginTest
LoginTest()
testCreateAccount_LoginValidation_AddTa
seleniumtest.LoginTest
testCreateAccount_LoginValidation_AddTask
Test Results
Output - SeleniumTest (test)
seleniumtest.LoginTest
Tests passed: 100.00 %
The test passed. (74.042 s)
New account created successfully
Example 1 Passed
Login successful
Example 2 Passed - Task added succ
Apr 20, 2026 12:27:39 PM org.openq
WARNING: Unable to find an exact m
Finished building SeleniumTest (test).
26:1
INS
12:28 PM
4/20/2026

TEST RESULTS

PASS/FAIL RESULTS:
ALL 28 TEST CASES PASSED SUCCESSFULLY, INDICATING THAT THE SYSTEM OPERATES CORRECTLY ACCORDING TO THE DEFINED REQUIREMENTS.

NUMBER OF PASSED TEST CASES: 28 (100%)
NUMBER OF FAILED TEST CASES: 0 (0%)

All functions worked correctly

CONCLUSION

- Selenium was effective for automated testing
- Helped save time and improve accuracy
- Main features were tested successfully
- More test cases can be added in the future

Thank you.